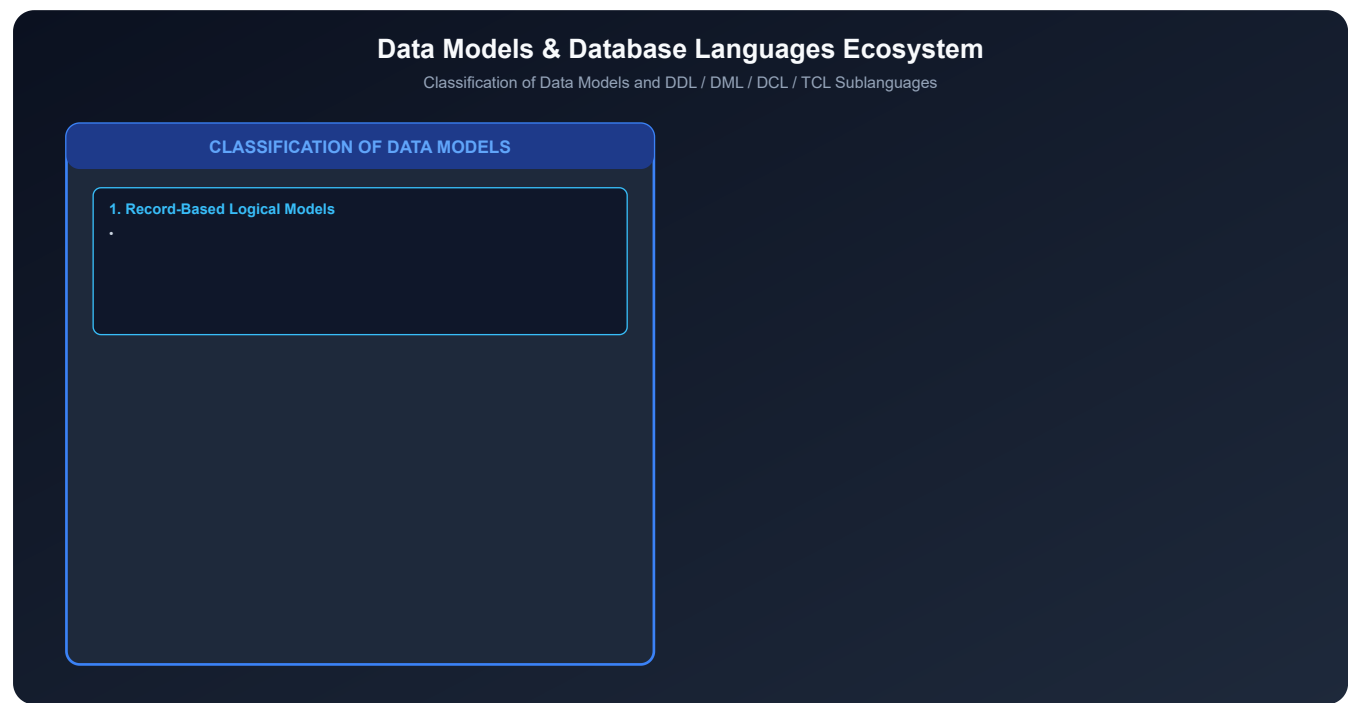


Module 04: Data Models and Languages

1. Data Models Classification



Relational Model: Data organized in 2D tables (relations). • **Network Model (CODASYL):** Records connected via pointers/graphs. • **Hierarchical Model:** Tree structure, parent-child segments (IMS). Standard across industry: Relational dominates. 2. Object-Based Data Models • **Object-Oriented (OODBMS):** Objects, classes, methods, inheritance. • **Object-Relational (ORDBMS):** Relational + complex user-defined types (PostgreSQL). 3. Semi-Structured Models (NoSQL) • JSON documents, BSON, XML trees (MongoDB, DynamoDB). Schema-on-read flexibility without predefined DDL tables. 4. Physical Data Models Low-level record formats, byte-offsets, block headers. DATABASE LANGUAGES (SQL SUBSYSTEMS) DDL (Data Definition Language) Defines schema structure & writes metadata to Data Dictionary. Commands: CREATE, ALTER, DROP, TRUNCATE, RENAME DML (Data Manipulation Language) Accesses and manipulates user data inside tables. • Procedural DML: User specifies WHAT and HOW (Relational Alg) • Non-Procedural (Declarative) DML: User specifies WHAT only (SQL) Commands: SELECT, INSERT, UPDATE, DELETE DCL (Data Control Language) Manages security privileges and access rights. Commands: GRANT, REVOKE TCL (Transaction Control Language) Controls transaction boundaries and atomicity rollback. Commands: COMMIT, ROLLBACK, SAVEPOINT

2. Procedural vs Non-Procedural DML (AKTU Essential)

Feature	Procedural DML	Non-Procedural (Declarative) DML
Query Specification	Specifies WHAT data is needed AND HOW to retrieve it	Specifies WHAT data is needed WITHOUT HOW
Execution Control	User/Programmer controls traversal algorithm	DBMS Query Optimizer decides optimal physical plan
Complexity	High programming overhead	Simple, intuitive, high-level

Primary Examples	Relational Algebra, Network/Hierarchical DML	SQL, Relational Calculus
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3. Database Sublanguages (DDL / DML / DCL / TCL)

- **DDL:** CREATE, ALTER, DROP, TRUNCATE (Updates System Catalog Metadata)
- **DML:** SELECT, INSERT, UPDATE, DELETE (Manipulates Table Tuples)
- **DCL:** GRANT, REVOKE (Access Control Privileges)
- **TCL:** COMMIT, ROLLBACK, SAVEPOINT (Transaction State Control)